

This dataset contains patient health information relating to the likelihood of stroke, containing upwards of 4000 authentic entries, as well as additional synthetic data created from the SMOTE process. It includes demographic details such as gender, age, work type, and residence. Medical indicators like hypertension, heart disease, glucose level, BMI and smoking status are also present. The target variable is “stroke”, indicating whether the patient had a stroke or not. While this data can be utilized in many different ways, our group opted to pursue a predictive model that can take an individual’s individual risks, combine and compare them to previous patients’ data, and provide a score for the patient describing their likelihood of experiencing a stroke given the parameters provided. This ability to identify a patient with a high risk of stroke would allow doctors to give priority to said patients and establish the amount of urgency and care that would be required in order to decrease the overall risk of stroke for the patient. As we learn more about machine learning, how it operates using data, and its limits, we may choose to alter slightly how we utilize the final results but overall, the direction should remain the same (using the data to highlight stroke probability).